

3.2 Self-Attention (Q, K, V)

Queries ask, keys are asked, values are mixed. Softmax weights over the sequence.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

Self-attention is a weighted sum of **values**, with weights from a comparison of **queries** to **keys**. Every token plays all three roles.

1. Q, K, V

For a sequence matrix $X \in \mathbb{R}^{T \times d}$,

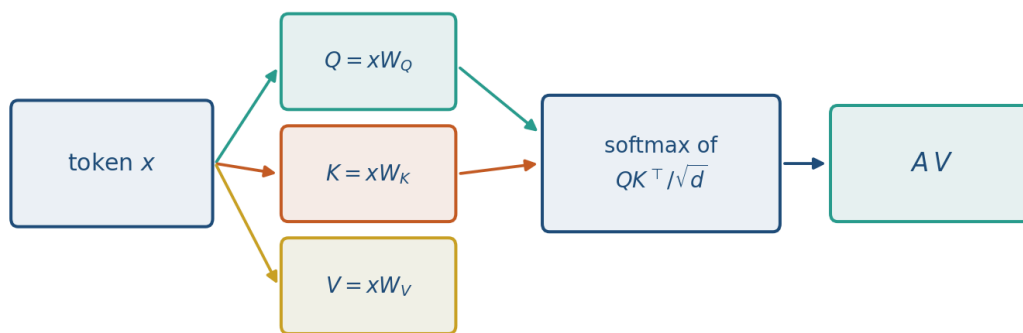
$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V.$$

Scores are scaled dot products, then a softmax over keys:

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right), \quad \text{Attention}(Q, K, V) = AV.$$

The $\sqrt{d_k}$ keeps the dots from saturating the softmax when d_k is large.

Query asks, key is asked, value is what you mix in

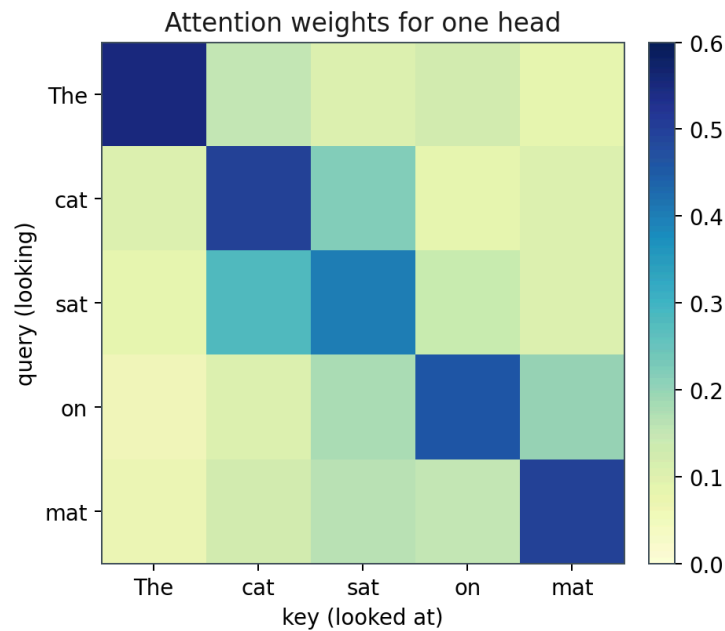


Query: what this token is looking for. **Key:** what another token looks like to be found. **Value:** what you actually add into the representation.

Row t of QK^T is "how much token t wants each position." Softmax makes that a distribution. Then you mix **values**, not keys. Mixing keys is a common lab bug.

2. A row of weights

Row t of A is a distribution over the sequence. Large mass on a name is "this pronoun just looked at that name."



Multi-head: several (W_Q, W_K, W_V) in parallel, concatenate, project. Different heads can track syntax, a name, or a comma. Lab 3 will compute one head by hand.

If two keys are equal, the softmax splits mass equally between them (given the same query). That is not a tie-break bug; it is the definition.

3. Masks

For next-token prediction you **mask** future keys (set scores to $-\infty$ before softmax). Otherwise the model cheats. BERT does not mask the future; it masks random tokens and reconstructs them (note 3.4).

Zero is the wrong mask fill-in: $\text{softmax}([1, 0])$ still puts mass on the “zero.” $-\infty$ becomes probability 0.

4. Teaching this note

30–40 minutes. Define Q, K, V in one sentence each, write $A = \text{softmax}(QK^T / \sqrt{d_k})$, then grind the 2×2 worked example on the board (including $\sqrt{d_k}$). Play **the QKV and softmax stretch** of 3Blue1Brown attention (about **4:00–16:00**). Lab 3 is $T = 4$; do not start that notebook until this arithmetic is done once by hand.

5. Worked example

$T = 2, d_k = 2$, skip W_Q and take scores $S = QK^T$ already:

$$S = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \quad \frac{S}{\sqrt{2}} \approx \begin{bmatrix} 1.41 & 0 \\ 0 & 1.41 \end{bmatrix}.$$

Row 1 softmax: $e^{1.41} \approx 4.10$, $e^0 = 1$, so $[4.10, 1] / 5.10 \approx [0.80, 0.20]$.

Row 2 is the same on the diagonal: $A \approx \begin{bmatrix} 0.80 & 0.20 \\ 0.20 & 0.80 \end{bmatrix}$.

If $V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then $AV \approx A$. Each output is 80% "self" and 20% "the other token."

Causal mask on row 1: set the future score to $-\infty$. Then row 1 softmax is $[1, 0]$: token 1 cannot look at token 2.

6. Where students get stuck

- Softmax over the wrong axis (columns instead of keys / last dim).
 - Dividing by d_k instead of $\sqrt{d_k}$, or skipping the scale.
 - Masking with 0 instead of $-\infty$.
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7. Video

Watch [3Blue1Brown: Attention in transformers, visually explained](#).

Pause when queries and keys meet as a grid of dots, when softmax turns a row into a distribution, and when values are mixed. That grid is Lab 3's heatmap.

8. Practice

1. If two tokens have identical keys, what does the softmax do to their weights?
2. Why divide by $\sqrt{d_k}$?
3. Write A for $T = 2$ with $Q = K = V = I$ (identity), no mask. What is AV ?
4. Softmax the row $[0, 2]$. Then softmax $[0, 2] / \sqrt{2}$. Did the larger entry gain or lose probability after scaling?
5. Scores $\begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$, causal mask (upper triangle $-\infty$). Write the masked score matrix, then A (row-wise softmax).